Electronic configuration

The general valence shell electronic configuration of alkali metals is ns^1, where 'n' represents the period number.

Table 5.2 Electronic configuration of alkali metals

Element	Symbol	Atomic No.	Electronic configuration
Lithium	Li	3	$[He]2s^1$
Sodium	Na	11	$[Ne]3s^1$
Potassium	K	19	$[Ar]4s^1$
Rubidium	Rb	37	$[Kr]5s^1$
Caesium	Cs	55	$[Xe]6s^1$
Francium	Fr	87	$[Rn]7s^1$

Common oxidation state

All these elements are highly electropositive in nature. They readily lose their valence electron to give monovalent cations (M^+). Alkali metals have only one oxidation state which is +1.

Atomic and ionic radii

Being the first element of each period, alkali metals have the largest atomic and ionic radii in their respective periods. On moving down the group, there is an increase in the number of shells and, therefore, atomic and ionic radii increase. The monovalent ions (M^+) are smaller than the respective parent atoms as expected.

Table 5.3 Physical properties of alkali metals

Physical property	Li	Na	K	Rb	Cs
Atomic radius / Metallic radius (Å)	1.52	1.86	2.27	2.48	2.65
Ionic radius (Å)	0.76	1.02	1.38	1.52	1.67
Melting point (^{0} C)	181	98	63	39	28.5
Boiling point (^{0} C)	1347	881	766	688	705
First ionization enthalpy (kJ mol^{-1})	520.2	495.8	418.8	403.0	375.7
Electronegativity (Paulings scale)	1.0	0.9	0.8	0.8	0.7
Density (g cm^{-3})	0.54	0.97	0.86	1.53	1.90
Standard potential E^0 for M^+/M (V)	-3.04	-2.71	-2.92	-2.93	-2.93
Hydration enthalpy (kJ mol^{-1})	-506	-406	-330	-310	-276

11th Std Chemistry 125-157.indd 127 06-01-2021 15:29:12